

Peltier Modules as Power Generators (TEG): A Solid-State Physics Analysis

While commercially recognized for active cooling via the Peltier effect, standard Bismuth Telluride (Bi_2Te_3) thermoelectric coolers (TECs) exhibit operational reversibility. When subjected to an externally applied thermal gradient, they function as Thermoelectric Generators (TEGs), directly converting sensible heat into electrical potential. This process operates independent of macroscopic moving parts, relying entirely on the fundamental quantum mechanics of charge carrier diffusion.

Thermoelectric Generation: The Kelvin Relations and Seebeck Effect

The core operational principle of a TEG is governed by the coupled thermoelectric phenomena, primarily the Seebeck effect, which describes the induction of a continuous electromotive force (EMF) under a temperature gradient (∇T). This is intrinsically linked to the Peltier and Thomson effects via the Kelvin (Thomson) relations.

Mathematically, the open-circuit voltage (V_{oc}) generated is the spatial integral of the temperature gradient multiplied by the material's temperature-dependent Seebeck coefficient (S):

$$V_{oc} = \int_{T_c}^{T_h} S_{AB}(T) dT \approx S_{pn} \Delta T$$

Where:

- T_h and T_c are the absolute temperatures of the hot and cold junctions, respectively.
- $S_{pn} = S_p - S_n$ is the relative Seebeck coefficient of the p-type and n-type semiconductor couple.
- ΔT is the macroscopic temperature difference across the module junctions.

Charge Carrier Dynamics

In heavily doped semiconductors, the thermal gradient severely alters the Fermi-Dirac statistical distribution. At the hot junction (T_h), elevated thermal energy excites majority carriers (electrons in n-type, holes in p-type) above the Fermi level, increasing their kinetic energy. This concentration of high-energy carriers drives isotropic diffusion toward the cold junction (T_c). The resulting charge accumulation at the cold junction creates an opposing internal electrostatic field ($\vec{E}$). At steady-state open-circuit conditions, the drift current induced by $\vec{E}$ perfectly balances the thermal diffusion current, manifesting as the measurable voltage V_{oc} .

The Thomson Effect

In a rigorous thermodynamic analysis, heat is also absorbed or evolved along the length of the semiconductor leg due to the Thomson effect, where a current I flowing through a thermal

gradient ∇T interacts with the Thomson coefficient (μ):

$$\dot{q}_\tau = \mu I \nabla T$$

While often neglected in simplified models, the Thomson heat significantly shifts the temperature profile from a linear Fourier distribution to a parabolic curve during high-current extraction.

Working Principle and Thermal Circuit Analysis

A standard TEG module consists of heavily doped semiconductor pellets arranged electrically in series (to boost voltage) and thermally in parallel (to minimize thermal resistance), sandwiched between two highly thermally conductive, electrically insulating ceramic substrates (e.g., Al_2O_3 or AlN).

To accurately model the thermal flux, the entire assembly must be treated as a **1D Thermal Resistance Network**:

$$R_{th_total} = R_{source} + R_{TIM1} + R_{TEG} + R_{TIM2} + R_{sink}$$

The effective temperature difference *actually* dropped across the active semiconductor pellets ($\Delta T_{pellets}$) is always less than the external temperature difference applied across the system due to parasitic thermal resistances (R_{TIM}).

1. The Hot Junction (T_h) and Heat Flux

The hot junction absorbs the input thermal flux (Q_h). The total heat absorbed dynamically changes based on the electrical load due to the Peltier heat absorbed at the hot junction and Joule heating:

$$Q_h = S_{pn} I T_h + K_{TEG} \Delta T - \frac{1}{2} I^2 R_{int}$$

Where I is the load current, K_{TEG} is the macroscopic thermal conductance of the module, and R_{int} is the module's internal electrical resistance. The $\frac{1}{2}$ term assumes Joule heating is distributed evenly and symmetrically conducts to both the hot and cold junctions.

2. The Cold Junction (T_c) and Heat Rejection

A high-performance heat sink is strictly required on the cold side. The heat sink must dissipate Q_c , which is the sum of the thermal conduction and the rejected electrical heat:

$$Q_c = S_{pn} I T_c + K_{TEG} \Delta T + \frac{1}{2} I^2 R_{int}$$

If the heat sink's thermal resistance (R_{sink}) is too high, T_c will rise as

$T_c = T_{ambient} + Q_c R_{sink}$, rapidly collapsing ΔT and halting power generation.

Energy Conversion and Material Science Considerations

The thermodynamic efficiency of a TEG is bottlenecked by the intrinsic material properties, defined by the dimensionless **Figure of Merit (ZT)**:

$$ZT = \frac{S^2 \sigma T}{\kappa_{total}}$$

Where:

- σ = Electrical conductivity
- $\kappa_{total} = \kappa_{lattice} + \kappa_{electronic}$ = Total thermal conductivity
- T = Absolute operating temperature

Decoupling Electron and Phonon Transport

An optimal TEG material requires a high Seebeck coefficient, high electrical conductivity, and *low* thermal conductivity. This presents a fundamental contradiction (the Wiedemann-Franz Law dictates that materials with high σ generally have high $\kappa_{electronic}$).

Modern thermoelectric engineering focuses on **Phonon Glass-Electron Crystal (PGEC)** paradigms. By introducing nanostructural defects, alloying, or grain boundary engineering, researchers scatter acoustic phonons (reducing $\kappa_{lattice}$) without significantly impeding the mean free path of charge carriers (maintaining σ).

Material Regimes

- **Low-Temperature ($< 200^\circ C$):** Bismuth Telluride (Bi_2Te_3) alloys heavily doped with Antimony (Sb) or Selenium (Se).
- **Mid-Temperature ($200^\circ C - 600^\circ C$):** Lead Telluride (PbTe) and Skutterudites (e.g., $CoSb_3$).
- **High-Temperature ($> 600^\circ C$):** Silicon-Germanium (SiGe) alloys, utilized extensively in Radioisotope Thermoelectric Generators (RTGs) for deep-space missions.

The maximum theoretical conversion efficiency (η_{max}) is bounded by the Carnot limit, scaled by the material's figure of merit:

$$\eta_{max} = \left(1 - \frac{T_c}{T_h}\right) \frac{\sqrt{1 + ZT_M} - 1}{\sqrt{1 + ZT_M} + \frac{T_c}{T_h}}$$

(Where T_M is the mean temperature across the module). Standard Bi_2Te_3 TEGs operate at roughly 4% – 6% efficiency, whereas advanced multi-stage modules can approach 8% – 10%.

Electrical Output and Maximum Power Point Tracking (MPPT)

The TEG acts as an active, non-ideal voltage source with a dynamic internal resistance R_{int} (which varies slightly with temperature). To extract maximum power (P_{max}), the external load

resistance (R_{load}) must strictly adhere to Jacobi's Law, where $R_{load} \approx R_{int}$.

Under perfect impedance matching, the maximum electrical power is given by:

$$P_{max} = \frac{V_{oc}^2}{4R_{int}} = \frac{S_{pn}^2 (\Delta T)^2}{4R_{int}}$$

Power Conditioning

Because V_{oc} fluctuates wildly with ambient thermal conditions (typically between $0.5V$ and $4V$), a direct connection to a static load is highly inefficient. Production-grade TEG systems utilize **Ultra-Low-Voltage DC-DC Boost Converters** equipped with **Maximum Power Point Tracking (MPPT)** microcontrollers.

Using algorithms like *Perturb and Observe (P&O)*, the converter continuously adjusts its input impedance. It measures the instantaneous V and I , alters the PWM duty cycle of the boost converter, and calculates if the resulting P increased or decreased, ensuring the system constantly tracks the P_{max} point despite varying thermal gradients.

Crucial Electro-Thermal Handling Tips and Failure Modes

1. **Minimization of Thermal Contact Resistance (R_{tc}):** Macroscopic surface roughness creates micro-voids of trapped air (an excellent thermal insulator). High-performance Thermal Interface Materials (TIMs)—such as micron-silver-particulate thermal paste, liquid metal, or phase-change materials—must be applied. Furthermore, modules must be clamped under specific compressive loads (typically $150 - 300 \text{ kPa}$) to minimize R_{tc} without fracturing the ceramic.
2. **Thermal Degradation Limits:** Standard TECs are manufactured using Bismuth-Tin (BiSn) solders with a eutectic melting point around $138^\circ C$. Exposing the ceramic substrate to high-grade thermal sources ($> 150^\circ C$) will cause rapid reflow of internal solder junctions and severe dopant diffusion, permanently destroying the module.
3. **Thermomechanical Fatigue (CTE Mismatch):** As the hot side expands and the cold side contracts, massive shear stresses are induced across the semiconductor pellets. This is governed by the Coefficient of Thermal Expansion (CTE) differential between the ceramic plates and the Telluride pellets. Over thousands of thermal cycles, this shear stress initiates micro-cracking in the pellets or at the solder joints, steadily increasing R_{int} until the module open-circuits. High-temperature TEGs use specialized buffer layers and flexible interconnects to mitigate CTE mismatch.